

ACTIVITY-2

Construct an AVL tree 70, 50, 30, 90, 80, 130, 120 & 30

insert 70

(70)

insert 50

(70)¹
(50)₀

insert

(30)

(70)²
(50)¹
(30)₀

→

(50)₀
(30)₀ (70)₀

insert 90

(50)⁻¹
(30)₀ (70)⁻¹
(90)₀

insert 80

(50)₀
(30)₀ (80)₀ (90)₀

(R-L rotation)

insert 130

(50)⁻²
(30)₀ (80)⁻¹
(70)₀ (90)₀ (130)₀

→ (80)₀
(50)₀ (90)⁻¹
(30)₀ (70)₀ (130)₀

insert 120

(80)₀
(50)₀ (120)₀
(30)₀ (70)₀ (90)₀ (130)₀

Delete 120

→ (80)₀
(50)₀ (130)₀
(30)₀ (70)₀ (90)₀

Delete 30

(80)₀
(50)₀ (130)₀
(70)₀ (90)₀

Q. AVL Tree - 16, 27, 9, 11, 36, 54, 81, 63, 72

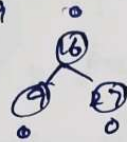
insert 16

(16)

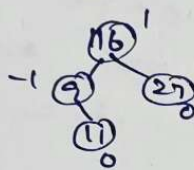
insert 27



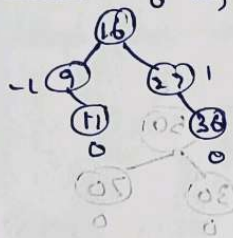
insert 9



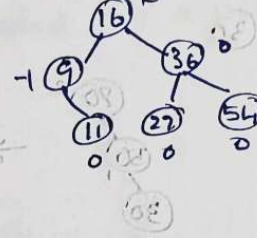
insert 11



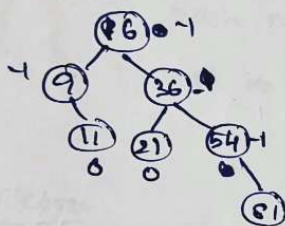
insert 36



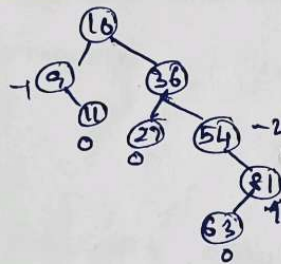
insert 54



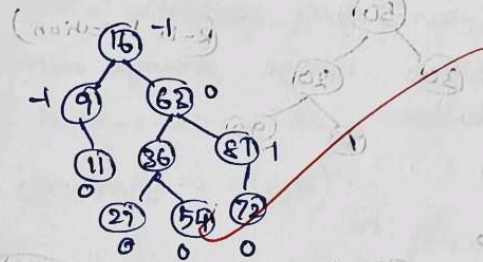
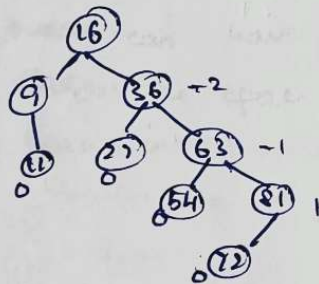
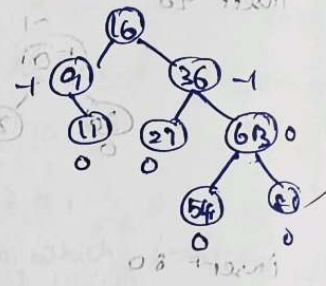
insert 81



insert 63



insert 72

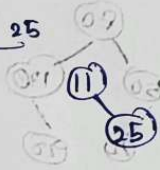


Q. Create a Binary Search tree using 11, 26, 10, 15, 39, 56, 84, 65, 78.

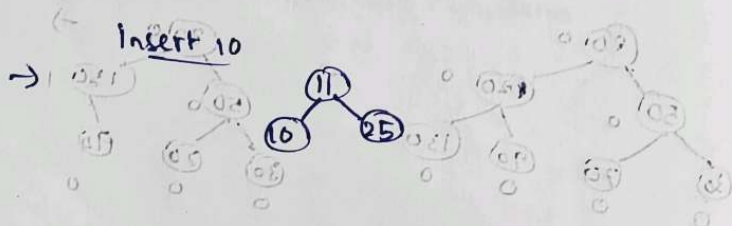
insert 11

(11)

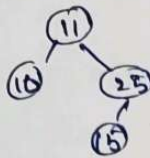
insert 25



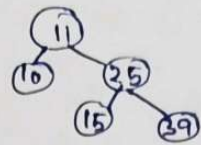
insert 10



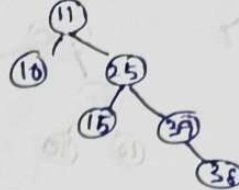
insert 15



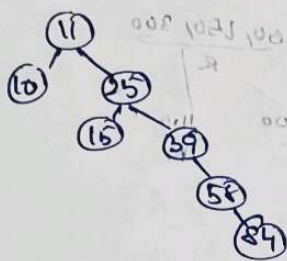
insert 39



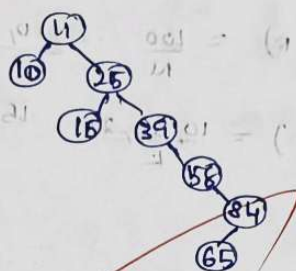
insert 58



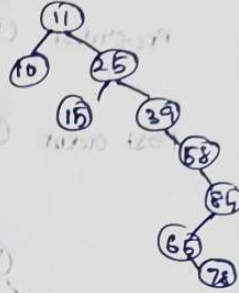
insert 84



insert 65



insert 28



4) Show binary tree that results after each of the integer key 9, 27, 50, 15, 2, 21, and 36.

insert 9



insert 27



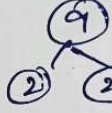
insert 50



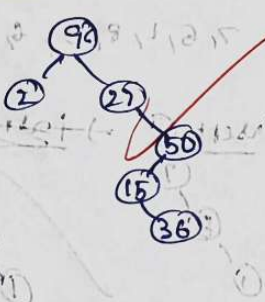
insert 15



insert 2



insert 36



Good
12/1/20

